

PHASE 7

PROJECT DOCUMENTATION PHASE

Project Introduction

ComicCraft is an automated AI comic story creator that leverages Google Gemini models and local diffusion technology to synthesize narratives and panel art.

Objectives

To provide an automated pipeline for generating, formatting, and exporting AI-driven comic books based on user input.

Technologies Used

Python, Google Gemini Pro, Google Gemini Flash, Local Diffusion, HTML/CSS, PDF Generation Libraries.

System Requirements

Requires Python, requisite pip packages (including local diffusion dependencies), and valid API keys stored in a `.env` file.

System Design

The system uses a modular architecture separating web routing, AI text generation, image synthesis, layout construction, and file exporting.

Implementation

Implemented via a Python backend using modular scripts (`gemini_pro.py`, `image_generator.py`, `layout_builder.py`, `exporters.py`) orchestrated by a central web application (`main.py`, `routes.py`).

Features

- Automated story generation via Gemini.
- Automated comic panel art generation via Local Diffusion.
- Automated PDF compilation and export.

Screenshots

 **ComicCraft AI**

Turn your idea into a five-panel comic.

Story Prompt

A brave fox exploring an enchanted forest

Main Character Name

Alex

Setting

forest

Story Tone

light-hearted

Art Style

anime

Generate Comic

Pipeline: Outline → Story & Dialogue → Illustrations → Layout → PDF

 **Your Comic**

Download Your Comic as PDF

Panel 1: The Beginning



Alex begins the adventure in the school.

Caption: The Beginning — Poetic moment.

Narration: Alex faces the moment described in this panel. Alex begins the adventure in the school.

► Image Prompt Reference

Panel 5: A New Beginning



Alex reaches the end of this adventure with a memorable lesson.

Caption: A New Beginning — Poetic moment.

Narration: Alex faces the moment described in this panel. Alex reaches the end of this adventure with a memorable lesson.

▼ Image Prompt Reference

Alex celebrating at the end of an adventure in a school, anime comic art, warm ending, cinematic

Create Another Comic

Advantages

- End-to-end automation of comic creation.
- Utilizes advanced LLMs for coherent storytelling.
- Cost-effective local image generation.

Conclusion

ComicCraft successfully demonstrates the integration of multiple distinct AI models—combining the textual reasoning capabilities of Google Gemini with the visual synthesis of local diffusion models—into a single, cohesive software pipeline capable of producing formatted digital media.

GitHub Repository: [REPO LINK](#)